ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PAPER 1 · PRACTICE B · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 1 · practice b.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable symbols.

## SECTION A

## QUESTION 1

**1.1** B ✓
**1.2** C ✓
**1.3** B ✓
**1.4** A ✓
**1.5** D ✓
**1.6** C ✓
**1.7** B ✓
**1.8** C ✓
**1.9** A ✓
**1.10** D ✓

(10 × 1) [10]

## QUESTION 2

**2.1** True ✓
**2.2** False ✓
**2.3** True ✓
**2.4** False ✓
**2.5** True ✓
**2.6** True ✓
**2.7** False ✓
**2.8** True ✓
**2.9** False ✓
**2.10** True ✓

(10 × 1) [10]

## QUESTION 3

**3.1** C ✓
**3.2** J ✓
**3.3** F ✓
**3.4** E ✓
**3.5** H ✓
**3.6** A ✓
**3.7** G ✓

**3.8** B ✓

**3.9** D ✓

**3.10** I ✓

(10 × 1) [10]

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1** Protons – positive, in the nucleus ✓; neutrons – no charge, in the nucleus ✓; electrons – negative, orbiting around the nucleus in shells ✓ (3)

**4.2** Metals have many free (loosely held) electrons in their outer shells ✓ that can move easily from atom to atom when a voltage is applied ✓. (2)

**4.3** (a) 4,7 kΩ ✓ (b) 25 mA ✓ (c) 2,2 × $10^{-6}$ F ✓ (d) 3 000 V ✓ (4)

**4.4** Any FIVE ✓✓✓✓✓: no food or drink; wear safety goggles when soldering or cutting; switch off before working on a circuit; never work with wet hands; check cables and tools for damage; return the soldering iron to its stand; tie back long hair / no loose clothing; know where the fire extinguisher and first-aid kit are; report injuries and faults. (5)

**[14]**

## QUESTION 5

**5.1.1** $R_T = 100 + 220 + 680 = 1\,000\ \Omega$ ✓ (1)

**5.1.2** $I = V \div R$ ✓ $= 12 \div 1\,000 = 0{,}012\ A\ (12\ mA)$ ✓ (2)

**5.1.3** $V = I \times R$ ✓ $= 0{,}012 \times 680 = 8{,}16\ V$ ✓ (2)

**5.2** $R_T = (6 \times 3) \div (6 + 3)$ ✓ $= 2\ \Omega$ ✓; $I = 12 \div 2 = 6\ A$ ✓; $I_{6\Omega} = 12 \div 6 = 2\ A$ ✓ (4)

**5.3.1** $R_P = (6 \times 12) \div (6 + 12) = 4\ \Omega$ ✓; $R_T = 4 + 4 = 8\ \Omega$ ✓ (2)

**5.3.2** $I = 24 \div 8 = 3\ A$ ✓ (1)

**5.3.3** $V_P = 3 \times 4 = 12\ V$ ✓ (1)

**5.3.4** $I_{12\Omega} = 12 \div 12 = 1\ A$ ✓ (1)

**5.4.1** $W = P \times t = 0{,}06\ kW \times 5 \times 20$ ✓ $= 6\ kWh$ ✓; $cost = 6 \times R2{,}50 = R15{,}00$ ✓ (3)

**5.4.2** $Q = I^2Rt$ ✓ $= 2^2 \times 10 \times 60$ ✓ $= 2\,400\ J$ ✓ (3)

**[20]**

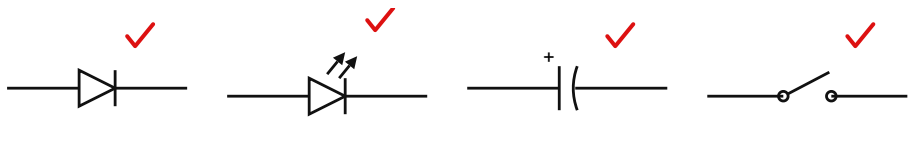
## QUESTION 6

**6.1** Yellow 4, violet 7 ✓, orange × 1 000 ✓ = 47 000 Ω (47 kΩ); gold = ±5% ✓ (3)

**6.2** (a) $C_T = 10 + 10 = 20\ \mu F$ ✓ (b) $1/C_T = 1/10 + 1/10$ ✓; $C_T = 5\ \mu F$ ✓ (3)

**6.3** Doping is adding a small amount of another element (impurity) to pure silicon to change how it conducts ✓. N-type: add an element with five outer electrons (e.g. phosphorus / arsenic) ✓, giving extra free electrons ✓. P-type: add an element with three outer electrons (e.g. boron / gallium), giving holes ✓. (4)

**6.4**




(a) diode (b) LED (c) electrolytic capacitor (d) switch

((a) triangle pointing to a bar 1; (b) diode symbol with two arrows pointing out 1; (c) one straight and one curved plate, with + marked 1; (d) a break in the line with a hinged contact 1)

(4)

**6.5** Without it the LED would draw too much current and burn out; the resistor limits the current ✓. $R = (V_s - V_{LED}) \div I$ ✓ $= (9 - 2) \div 0{,}02$ ✓ $= 350\ \Omega$ ✓ (4)

**[18]**

## QUESTION 7

**7.1** (a) 45 = 32 + 8 + 4 + 1 ✓ → 101101 ✓ (b) 128 + 64 + 16 + 4 + 2 ✓ = 214 ✓ (c) 214 ÷ 16 = 13 remainder 6 ✓ → D6 ✓ (6)

**7.2**

| A | B | OR | NAND |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 |

(OR column 2, NAND column 2)

(4)

**7.3.1** $Y = A \cdot B + C$ ✓✓ (2)

**7.3.2** 8 rows ($2^3$) ✓ (1)

**7.3.3** 1 (C alone makes Y = 1) ✓ (1)

**7.4** Analogue: changes smoothly and can have any value in a range ✓, e.g. a temperature sensor voltage, a microphone signal ✓. Digital: has only two levels, 0 and 1 (OFF and ON) ✓, e.g. a push button, the output of a logic gate ✓. (4)

**[18]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**